



How Well Do Pickleball Rating Systems Reflect Skill? Evidence From Collegiate Data

Introduction

Competitive pickleball relies on rating systems to estimate player skill and ensure fair matchups, yet there is limited empirical evidence evaluating their accuracy. This study examines whether widely used ratings reflect true competitive performance by analyzing collegiate match data. We compare DUPR and NCPA ratings based on their ability to predict match outcomes and investigate how rating structure and aggregation in doubles influence predictive accuracy.

Methodology

- **Data:**
 - ~4,000+ collegiate matches with DUPR ratings
 - Subset with overlapping NCPA ratings
- **Prediction Framework:**
 - Construct team ratings from player ratings
 - Predict match outcomes using rating differences
 - Evaluate accuracy vs baseline win rate
- **Calibration Analysis:**
 - Compare win probability vs rating difference
 - Assess whether ratings produce monotonic, interpretable relationships

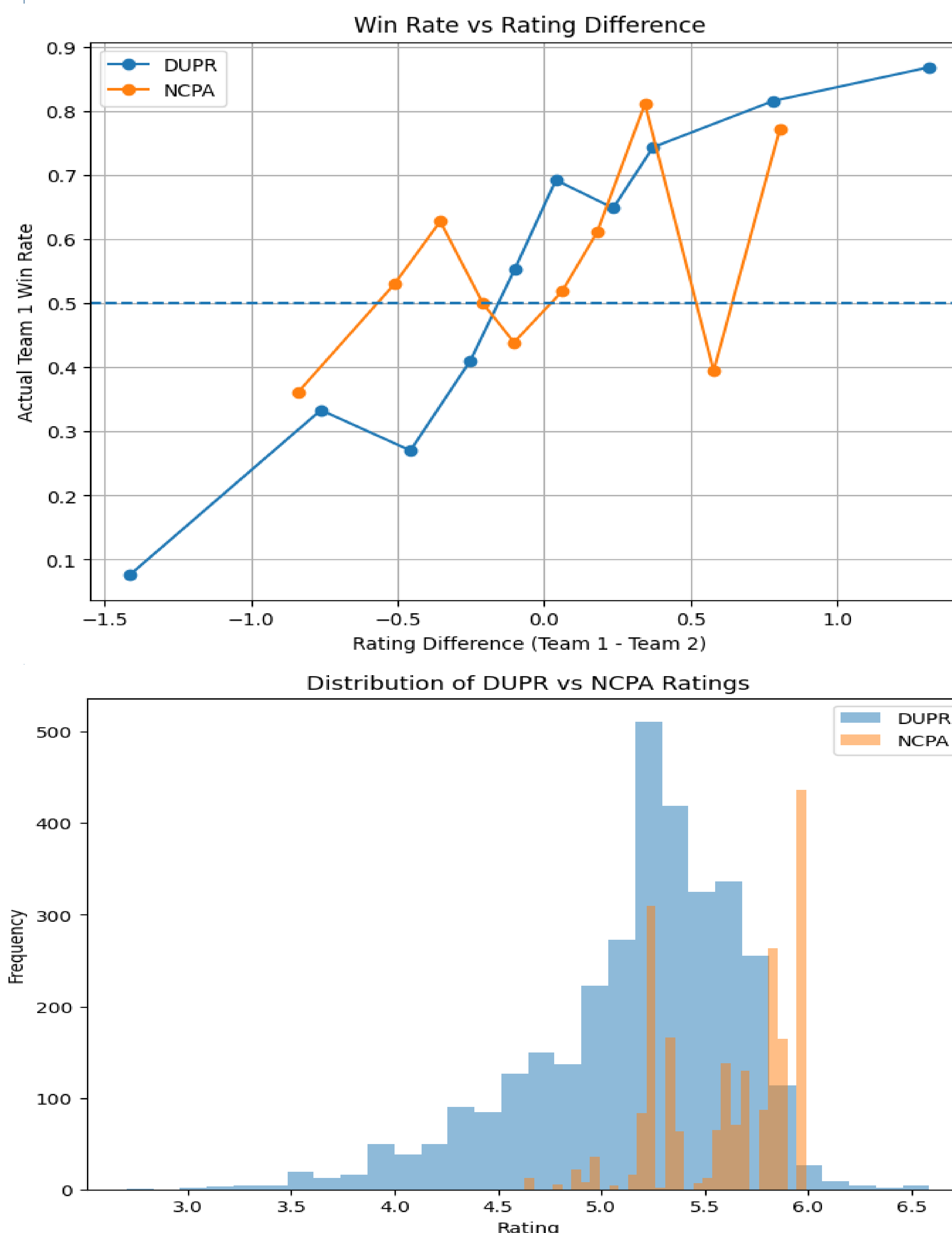
Doubles Modeling:

Test alternative aggregation methods (weighted vs equal)

DUPR vs NCPA

- DUPR vs NCPA
 - Prediction accuracy: ~82% vs ~48%
 - Relationship: Monotonic vs No Consistent Relationship
- Overall, DUPR drastically outperformed NCPA ratings in predicting match outcome

Key Insight: NCPA ratings are tightly clustered, limiting skill differentiation. DUPR's greater spread enables accurate outcome prediction



Weighted Doubles Ratings

Team Rating:

$$\alpha(\text{stronger}) + (1 - \alpha)(\text{weaker}), \alpha \in [0, 1]$$

- Result:
 - Prediction accuracy maximized at $\alpha = 0.5$
 - Equal weighting achieves ~82% accuracy
- Interpretation
 - Both players contribute equally to match outcomes
 - Overweighting either player degrades predictions

Key Insight: DUPR ratings already capture individual skill effectively. Alternative weighting schemes provide no meaningful improvement.

